Random Walks & Gambler's Ruin

Chapter 4.3 — Study Notes: Learning by Example

Jane Street National Probability Challenge · Subject 4: Stochastic Processes

How to use these notes

Every example has a **blue problem box**, a **green step-by-step solution**, and usually an **orange analogy**. Purple boxes hold formulas, teal boxes hold speed tips, and red boxes warn about traps.

Suggested workflow: (1) Read the Toolbox (Part A) once. (2) For each example, cover the solution and try the problem for 3–5 minutes. (3) Redo the examples marked "Mosteller" from memory a day later. (4) Finish with the rapid-fire drills in Part G.

Part	Theme	Examples
A	The Toolbox (formulas and pictures)	—
B	Gambler's Ruin: fair, biased, duration, martingales	1–8
C	One-sided barriers, drift and the cliff-hanger	9–13
D	Reflecting barriers, cycles, visit counts	14–17
E	Path counting: reflection, Catalan, ballot, returns	18–25
F	Higher dimensions (2D/3D)	26–27
G	Rapid-fire drills and one-page cheat sheet	—

Part A The Toolbox

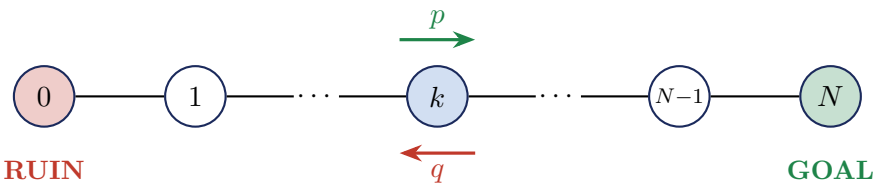
A.1 Setup and notation

Gambler's Ruin — the standard model

A walker (gambler) has fortune $k \in \{0, 1, \ldots, N\}$. Each round: $+1$ with probability p, -1 with probability $q = 1 - p$. The states 0 (ruin) and N (goal) are *absorbing*. Let

$$r := \frac{q}{p}, \qquad W_k := \mathbb{P}(\text{reach } N \text{ before } 0 \mid \text{start at } k), \qquad D_k := \mathbb{E}[\text{number of rounds} \mid \text{start at } k].$$

First-step analysis gives $W_k = pW_{k+1} + qW_{k-1}$ with $W_0 = 0$, $W_N = 1$, and $D_k = 1 + pD_{k+1} + qD_{k-1}$ with $D_0 = D_N = 0$.



The six formulas you must know cold

1. **Win probability.** Fair ($p = q$): $W_k = \dfrac{k}{N}$. Biased: $W_k = \dfrac{1 - r^k}{1 - r^N}$. Ruin probability is $1 - W_k = \dfrac{r^k - r^N}{1 - r^N}$.
2. **Duration.** Fair: $D_k = k(N - k)$. Biased: $D_k = \dfrac{N W_k - k}{p - q}$ (this is Wald's identity $\mathbb{E}[S_T] = k + (p - q)\mathbb{E}[T]$).
3. **Infinite bank** (let $N \to \infty$): $\mathbb{P}(\text{ever ruined from } k) = \min\{1, (q/p)^k\}$. For $p \le \frac{1}{2}$ ruin is certain.
4. **Simple walk** $S_n = X_1 + \cdots + X_n$, $X_i = \pm 1$ w.p. $\frac{1}{2}$: $\mathbb{P}(S_n = j) = \binom{n}{(n+j)/2} 2^{-n}$ and $u_{2n} := \mathbb{P}(S_{2n} = 0) = \binom{2n}{n} 4^{-n} \approx \frac{1}{\sqrt{\pi n}}$.
5. **Reflection principle.** Paths from 0 to $e < m$ that touch level $m > 0$ are in bijection with paths from 0 to $2m - e$. Hence $\mathbb{P}(\max_{i \le n} S_i \ge m) = 2\mathbb{P}(S_n > m) + \mathbb{P}(S_n = m)$.
6. **Ballot / Catalan.** If $a > b$: $\mathbb{P}(\text{A strictly ahead throughout}) = \frac{a-b}{a+b}$, and $\mathbb{P}(\text{A never behind}) = \frac{a-b+1}{a+1}$. Balanced paths of length $2n$ never going below 0: $C_n = \frac{1}{n+1}\binom{2n}{n}$.

Analogy – make it intuitive

Five pictures to carry in your head.

- **The drunk on a pier.** Sea on the left (0), a waiting taxi on the right (N). He staggers randomly. Gambler's ruin asks which end he reaches first.
- **Fair game = seesaw.** In a fair game the *expected fortune never changes*, like a seesaw's centre of mass. Final expectation $N \cdot W_k + 0 \cdot (1 - W_k)$ must equal the start k, so $W_k = k/N$ (the chance is the fraction of the way to the goal).
- **Bias = a tilted table.** A marble on a slightly tilted table drifts to one wall. The longer the table (bigger N or k), the more the tilt compounds, which is why r^k is exponential.
- **Reflection = a mirror.** Every path that touches the barrier has a ghost twin in the mirror that carries on upward. Counting mirror twins counts the bad paths.
- **Recurrence = Kakutani's quip.** "A drunk man will eventually find his way home, but a drunk bird may get lost forever." (1D and 2D walks return, 3D walks may not.)

Part B Gambler's Ruin: Fair, Biased, Duration and Martingales

Example 1 | The rat in the corridor (fair game)

A rat starts at position 3 of a corridor with positions $0, 1, \ldots, 10$. Poisoned cheese is at 0 and safe food at 10. Every second it moves one step left or right with probability $\frac{1}{2}$ each, stopping at 0 or 10.

(a) What is the probability it reaches the food?

(b) What is the expected number of seconds until it stops?

Step-by-step solution

Step 1. Let W_k be the chance of reaching 10 from k. Conditioning on the first move gives $W_k = \frac{1}{2}W_{k-1} + \frac{1}{2}W_{k+1}$, so $W_{k+1} - W_k = W_k - W_{k-1}$. Increments are constant, so W_k is *linear* in k.

Step 2. With $W_0 = 0$ and $W_{10} = 1$, the line is $W_k = k/10$. Hence $W_3 = 3/10$.

Step 3. For the time, $D_k = 1 + \frac{1}{2}D_{k-1} + \frac{1}{2}D_{k+1}$, i.e. $D_{k+1} - 2D_k + D_{k-1} = -2$. A constant second difference of -2 means D_k is a quadratic with $D_0 = D_{10} = 0$, so $D_k = k(10 - k)$. Therefore $D_3 = 3 \cdot 7 = 21$.

Answer. (a) $\frac{3}{10}$. (b) 21 seconds.

Analogy – make it intuitive

Picture a taut string from poison to food with a bead at position 3. In a fair game, probability is *distance fraction*: the bead is 30% of the way to the food, so it wins 30% of the time. The expected time $k(N - k)$ is the *area of the rectangle* formed by the distances to the two walls.

Example 2 | Poker night with unequal stacks

Alice has \$2 and Bob has \$8. Each hand is a fair coin flip for \$1. They play until someone is broke. Find (a) the probability Alice takes everything, (b) the expected number of hands, (c) Alice's expected final wealth.

Step-by-step solution

Step 1. Total money is $N = 10$ and Alice's start is $k = 2$, so the walk is fair with $k = 2$, $N = 10$.

Step 2. $\mathbb{P}(\text{Alice wins all}) = k/N = 2/10 = 0.2$.

Step 3. $\mathbb{E}[\text{hands}] = k(N - k) = 2 \cdot 8 = 16$.

Step 4. Final wealth is 10 w.p. 0.2 and 0 w.p. 0.8, so $\mathbb{E} = 10(0.2) = 2$, which equals her starting stake: the game is fair.

Answer. (a) 0.2 (b) 16 hands (c) \$2.

Analogy – make it intuitive

A seesaw: Alice is a light child (weight 2) and Bob a heavy adult (weight 8). The pivot (the "expected fortune") must stay where it is, so the lighter child hits the ground rarely: $2/(2+8)$ of the time.

Competition tip

Fair-game shortcut: win probability = (your stack)/(total money), and expected duration = (your stack)×(opponent's stack).

Example 3 | A biased game with a symmetric target

Each round you win \$1 w.p. 0.6 and lose \$1 w.p. 0.4. You start with \$5 and stop at \$10 or \$0. What is $\mathbb{P}$(reach \$10)?

Step-by-step solution

Step 1. $p = 0.6$, $q = 0.4$, so $r = q/p = 2/3$, with $k = 5$, $N = 10$.

Step 2. Formula: $W_5 = \dfrac{1 - r^5}{1 - r^{10}}$.

Step 3. Since $1 - r^{10} = (1 - r^5)(1 + r^5)$, this simplifies to $W_5 = \dfrac{1}{1 + r^5}$.

Step 4. $r^5 = (2/3)^5 = 32/243$, so $W_5 = \dfrac{1}{1 + 32/243} = \dfrac{243}{275} \approx 0.884$.

Answer. $\dfrac{243}{275} \approx 0.884$.

Analogy – make it intuitive

A marble on a table tilted only slightly (60/40), but the table is 10 units long. Even a small tilt compounds over many steps, so a 10-point edge per round becomes an 88% chance of reaching the far wall.

Competition tip

Halfway trick: if $N = 2k$, then $W_k = \dfrac{1}{1 + r^k}$. It saves a lot of arithmetic.

Example 4 | Mosteller #36 — Gambler's Ruin with $M = \$1$ and $N = \$2$

Player M has \$1 and player N has \$2. Each play moves \$1 from one to the other, and M wins each play with probability $\frac{2}{3}$. They play until one is bankrupt. Find $\mathbb{P}(M$ wins), the expected number of plays, and M's expected net gain.

Step-by-step solution

Step 1. Track M's fortune: states $0, 1, 2, 3$ with $p = \frac{2}{3}$, $q = \frac{1}{3}$, absorbing at 0 and 3.

Step 2. First-step equations for the win probability: $W_1 = pW_2$ and $W_2 = p + qW_1$.

Step 3. Substituting, $W_1 = p^2 + pq\,W_1$, so $W_1 = \dfrac{p^2}{1 - pq} = \dfrac{4/9}{1 - 2/9} = \dfrac{4/9}{7/9} = \dfrac{4}{7}$.

Step 4. (Check with the formula: $r = \frac{1}{2}$, $W_1 = \dfrac{1 - 1/2}{1 - 1/8} = \dfrac{1/2}{7/8} = \dfrac{4}{7}$.)

Step 5. Duration: $D_1 = 1 + pD_2$ and $D_2 = 1 + qD_1$. Then $D_1 = 1 + \frac{2}{3}(1 + \frac{1}{3}D_1)$, so $\frac{7}{9}D_1 = \frac{5}{3}$ and $D_1 = \frac{15}{7} \approx 2.14$.

Step 6. Net gain: M gains \$2 w.p. $\frac{4}{7}$ and loses \$1 w.p. $\frac{3}{7}$, so $\mathbb{E} = \frac{8}{7} - \frac{3}{7} = \frac{5}{7}$ dollars.

Answer. $\mathbb{P}(M$ wins$) = \frac{4}{7}$; $\mathbb{E}$[plays] $= \frac{15}{7}$; $\mathbb{E}$[gain] $= \frac{5}{7}$ dollars.

Analogy – make it intuitive

M is half as rich as N, but each play favours him 2:1. The moral from the text: it is better to be a better player than to be twice as wealthy.

Example 5 | Mosteller #37 — Bold play vs. cautious play

At roulette (American wheel: win probability $\frac{18}{38}$ on "evens"), a man with \$20 needs \$40. Strategy 1 (bold): bet all \$20 at once. Strategy 2 (cautious): bet \$1 at a time until he hits \$40 or \$0. Compare the probabilities of success.

Step-by-step solution

Step 1. *Bold:* one bet decides everything, so $\mathbb{P} = \frac{18}{38} = \frac{9}{19} \approx 0.474$.

Step 2. *Cautious:* gambler's ruin with $k = 20$, $N = 40$, $p = \frac{18}{38}$, $q = \frac{20}{38}$, so $r = \frac{20}{18} = \frac{10}{9}$.

Step 3. Using the halfway trick ($N = 2k$): $W_{20} = \dfrac{1}{1 + r^{20}}$.

Step 4. $\ln(10/9) \approx 0.10536$, so $r^{20} = e^{2.107} \approx 8.23$, giving $W_{20} = \dfrac{1}{9.23} \approx 0.108$.

Answer. Bold: ≈ 0.474. Cautious: ≈ 0.108. Bold play is more than four times better.

Analogy – make it intuitive

Think of the house edge as a tiny *tax per bet*. Cautious play makes about 400 small bets, paying the tax each time. Bold play pays it once. To beat an unfavourable casino, get in, take one big swing, and get out.

Example 6 | How bet size changes the odds of doubling up

Same American roulette ($p = \frac{18}{38}$ on even-money bets). You hold \$100 and want \$200. Compute the success probability if each bet is \$100, \$20, \$10, or \$1.

Step-by-step solution

Step 1. Express everything in bet-units: $N = 200$/bet and $k = 100$/bet, so $N = 2k$ and $W = \dfrac{1}{1 + r^k}$ with $r = \dfrac{10}{9}$.

Step 2. Compute r^k for each case: $r^1 = 1.111$, $r^5 = 1.694$, $r^{10} = 2.868$, $r^{100} = e^{10.536} \approx 3.76 \times 10^4$.

Step 3. Tabulate:

Bet size	k	N	$\mathbb{P}$(double up)
\$100	1	2	0.474
\$20	5	10	0.371
\$10	10	20	0.259
\$1	100	200	2.7×10^{-5}

Answer. 0.474, 0.371, 0.259, 2.7×10^{-5} respectively.

Common trap

Splitting a bet into many small bets does *not* reduce your risk against a negative-edge game. By the law of large numbers, you converge to the average loss per bet ($-\frac{2}{38}$ per dollar here). Small bets are only "safe" when the edge is in your favour.

Example 7 | Ties are just pauses

A casino game: you win \$1 w.p. 0.45, lose \$1 w.p. 0.45, and nothing happens (a tie) w.p. 0.10. You start with \$4 and play until you reach \$10 or \$0. Find the win probability and the expected number of rounds, *including* ties.

Step-by-step solution

Step 1. Ties do not move the walk, so condition on a non-tie round. Given a non-tie, the win probability is $\frac{0.45}{0.90} = \frac{1}{2}$. So the embedded walk is fair with $k = 4$, $N = 10$.

Step 2. $\mathbb{P}(\text{reach \$10}) = 4/10 = 0.4$.

Step 3. The embedded fair walk needs $k(N-k) = 4 \cdot 6 = 24$ non-tie rounds on average.

Step 4. Each round is a non-tie w.p. 0.9, so total rounds $= \frac{24}{0.9} = \frac{80}{3} \approx 26.7$ (each effective step costs on average $1/0.9$ rounds).

Answer. 0.4 and $\frac{80}{3} \approx 26.7$ rounds.

Analogy – make it intuitive

A frog that sometimes *naps* instead of hopping. Naps don't change where it ends up, just how long the trip takes. In general, duration $= k(N-k)/\sigma^2$, where σ^2 is the variance of one round (0.9 here).

Example 8 | Derive the ruin formulas with martingales (optional stopping)

Using only "a martingale's expectation is conserved at the stopping time T", derive (a) $W_k = k/N$ and $\mathbb{E}[T] = k(N-k)$ for the fair walk, and (b) the biased formula $W_k = \frac{1-r^k}{1-r^N}$.

Step-by-step solution

Step 1. *Fair walk.* S_n and $S_n^2 - n$ are both martingales (since $\mathbb{E}[X^2] = 1$). The stopped values are bounded by N, so optional stopping applies.

Step 2. $\mathbb{E}[S_T] = k$. But $S_T = N$ w.p. W_k and 0 otherwise, so $NW_k = k$ and $W_k = k/N$.

Step 3. $\mathbb{E}[S_T^2 - T] = k^2$. Now $\mathbb{E}[S_T^2] = N^2 W_k = Nk$, so $\mathbb{E}[T] = Nk - k^2 = k(N-k)$.

Step 4. *Biased walk.* $M_n = r^{S_n}$ with $r = q/p$ is a martingale because $\mathbb{E}[r^X] = p \cdot \frac{q}{p} + q \cdot \frac{p}{q} = q + p = 1$.

Step 5. Optional stopping: $r^k = \mathbb{E}[r^{S_T}] = (1 - W_k) \cdot 1 + W_k \, r^N$. Solving, $W_k = \dfrac{1 - r^k}{1 - r^N}$.

Answer. The standard formulas, derived without solving any recurrence.

Analogy – make it intuitive

A martingale is a *conserved quantity*, like energy in physics. Find something that stays constant in expectation (S_n, or r^{S_n}), evaluate it at the two possible exits, and match the books. It is the

fastest route to many competition answers.

Example 9 | Expected duration of a biased game

$p = 0.6$, you start with \$3 and stop at \$5 or \$0. Find the expected number of rounds.

Step-by-step solution

Step 1. $r = \frac{2}{3}$, $k = 3$, $N = 5$. First find the win probability: $W_3 = \dfrac{1 - r^3}{1 - r^5} = \dfrac{19/27}{211/243} = \dfrac{171}{211} \approx 0.810$.

Step 2. Wald's identity: $\mathbb{E}[S_T] = k + (p - q)\mathbb{E}[T]$, and $\mathbb{E}[S_T] = N\,W_k = 5 \cdot \frac{171}{211} = \frac{855}{211}$.

Step 3. So $\mathbb{E}[T] = \dfrac{NW_k - k}{p - q} = \dfrac{\frac{855}{211} - 3}{0.2} = \dfrac{222/211}{1/5} = \dfrac{1110}{211} \approx 5.26$.

Answer. $\dfrac{1110}{211} \approx 5.26$ rounds.

Competition tip

Duration in a biased game in three seconds: compute W_k, then $D = \dfrac{NW_k - k}{p - q}$. No second recurrence needed.

Part C One-Sided Barriers, Drift and the Cliff-Hanger

Example 10 | Mosteller #35 — The cliff-hanger

A drunk man stands one step from a cliff edge. Each step he moves *away* from the cliff w.p. $\frac{2}{3}$ and *toward* it w.p. $\frac{1}{3}$. (a) What is his chance of escaping the cliff forever? (b) What if he starts 3 steps from the edge?

Step-by-step solution

Step 1. Let x be the distance from the cliff; stepping to $x = 0$ means falling. This is gambler's ruin with an infinite bank: $p = \frac{2}{3}$ up, $q = \frac{1}{3}$ down.

Step 2. Let P_1 be the probability of eventually hitting 0 from 1. Condition on the first step: $P_1 = q + pP_2$. To get from 2 to 0 one must first get to 1 and then to 0, so $P_2 = P_1^2$. Hence

$$P_1 = q + pP_1^2 \iff pP_1^2 - P_1 + q = 0 \iff (P_1 - 1)(pP_1 - q) = 0,$$

so $P_1 = 1$ or $P_1 = q/p$.

Step 3. For $p > \frac{1}{2}$ the correct root is the smaller one, $P_1 = q/p = \frac{1/3}{2/3} = \frac{1}{2}$. (The root 1 is the one that applies when $p \le \frac{1}{2}$.)

Step 4. From distance m: $P_m = (q/p)^m$, so for $m = 3$, $P_3 = \frac{1}{8}$ and the escape probability is $\frac{7}{8}$.

Step 5. Sanity check: the cliff probability after 1, 3, 5 steps is $\frac{1}{3}$, $\frac{11}{27}$, $\frac{107}{243} \approx 0.44$, creeping up toward 0.5.

Answer. (a) Escape probability $\frac{1}{2}$. (b) $1 - \frac{1}{8} = \frac{7}{8}$.

Analogy – make it intuitive

A boat drifting in a gentle current away from a waterfall. A lucky streak moves you far from the edge. Each additional step of distance multiplies your danger by the same factor q/p, like stacking independent safety nets.

Example 11 | The expected highest peak of a downhill walk

A walker starts at 0 and moves $+1$ w.p. $p = 0.45$ and -1 w.p. $q = 0.55$ forever. (a) What is the probability that he *ever* reaches $+3$? (b) What is the expected value of his all-time maximum M?

Step-by-step solution

Step 1. Reaching $+a$ for a walk with up-probability $p < \frac{1}{2}$ is the mirror image of the cliff problem with roles swapped: $\mathbb{P}(\text{ever reach } +1) = \dfrac{p}{q} = \dfrac{9}{11}$.

Step 2. Levels are climbed one at a time, independently: $\mathbb{P}(M \geq a) = (p/q)^a$. For $a = 3$: $\left(\frac{9}{11}\right)^3 = \frac{729}{1331} \approx 0.548$.

Step 3. $\mathbb{E}[M] = \sum_{a\geq 1} \mathbb{P}(M \geq a) = \sum_{a\geq 1} \left(\frac{p}{q}\right)^a = \dfrac{p/q}{1 - p/q} = \dfrac{p}{q-p} = \dfrac{0.45}{0.10} = 4.5.$

Answer. (a) $\frac{729}{1331} \approx 0.548$. (b) $\mathbb{E}[M] = 4.5$.

Analogy – make it intuitive

Hiking on a gentle downward slope. Every peak you ever climb is like a "record", and each record is $\frac{9}{11}$ as likely as the one before. The records form a geometric ladder, so the average peak height is the sum of that geometric series.

Example 12 | Uphill walk: ruin chance and time to reach a target

$p = 0.6$ up, $q = 0.4$ down, no upper limit. (a) Starting with \$3, what is the probability of eventually going broke? (b) Starting at 0, what is the expected number of steps to first reach $+10$?

Step-by-step solution

Step 1. (a) Infinite-bank formula: $\mathbb{P}(\text{ruin}) = (q/p)^k = \left(\frac{2}{3}\right)^3 = \frac{8}{27} \approx 0.296$.

Step 2. (b) Let T be the first time the walk hits $+10$. For $p > \frac{1}{2}$ this has finite mean, and at time T the position is exactly 10.

Step 3. Wald: $\mathbb{E}[S_T] = (p - q)\mathbb{E}[T]$, i.e. $10 = 0.2\,\mathbb{E}[T]$, so $\mathbb{E}[T] = 50$.

Answer. (a) $\frac{8}{27} \approx 0.296$. (b) 50 steps.

Competition tip

For drift $\mu = p - q \neq 0$, the time to climb a levels is a/μ ("distance divided by speed"). Use it whenever the target is on the side the drift pushes toward.

Example 13 | The fair walk paradox

A fair coin game: you win or lose \$1 per flip. You play until you are \$1 ahead of where you started. (a) Is it certain you will eventually be ahead? (b) What is the expected number of flips?

Step-by-step solution

Step 1. (a) Let P_1 be the chance of ever reaching $+1$. The cliff-hanger equation with $p = \frac{1}{2}$ is $P_1 = \frac{1}{2} + \frac{1}{2}P_1^2$, i.e. $(P_1 - 1)^2 = 0$, so $P_1 = 1$. It is certain.

Step 2. (b) Let $E = \mathbb{E}[T]$. After the first flip: with prob. $\frac{1}{2}$ we are done in 1 flip; with prob. $\frac{1}{2}$ we are at -1 and must climb two levels, costing two independent copies of T. So

$$E = \tfrac{1}{2} \cdot 1 + \tfrac{1}{2}\,(1 + 2E) = 1 + E,$$

which has no finite solution. Hence $E = \infty$.

Step 3. Cross-check: stopping at $+1$ *or* $-L$ takes $1 \cdot L = L$ flips on average, which is a lower bound for T for every L. So $\mathbb{E}[T] \geq L$ for all L.

Answer. (a) Yes, with probability 1. (b) The expected time is infinite.

Common trap

"Certain to win eventually" does *not* mean "quick", and it does not mean "safe". To bank \$1 you may need to dip arbitrarily far below zero first. This is why the doubling (martingale) betting system fails: it needs an infinite bankroll.

Example 14 | Return probability of a biased walk

A walk starts at 0 with $p = 0.6$ of $+1$ and $q = 0.4$ of -1. (a) What is the probability it ever returns to 0? (b) What is the expected number of visits to 0, counting time 0?

Step-by-step solution

Step 1. Condition on the first step. Up (w.p. 0.6): from $+1$ it must come back down to 0, which has probability $q/p = \frac{2}{3}$.

Step 2. Down (w.p. 0.4): from -1 it must climb one level to 0. With upward drift this is certain: probability 1.

Step 3. $\mathbb{P}(\text{return}) = 0.6 \cdot \frac{2}{3} + 0.4 \cdot 1 = 0.4 + 0.4 = 0.8$. (In general, $2\min(p, q)$.)

Step 4. The number of visits (including the start) is geometric: each visit is followed by another with probability 0.8, so $\mathbb{E}[\text{visits}] = \dfrac{1}{1 - 0.8} = 5$. (In general, $\frac{1}{|p-q|}$.)

Answer. (a) 0.8. (b) 5 visits.

Part D Reflecting Barriers, Cycles and Visit Counts

Example 15 | The frog on a ladder (reflecting barrier)

A frog sits on lily pads numbered $0, 1, \ldots, 10$. On pads $1, \ldots, 9$ it hops left or right with probability $\frac{1}{2}$ each. On pad 0 it is pushed to pad 1 with probability 1 (a wall). It stops on reaching pad 10. Starting from pad 0, what is the expected number of hops?

Step-by-step solution

Step 1. Idea: unfold the wall. If S_n is an ordinary fair walk on $\mathbb{Z}$, then $|S_n|$ behaves exactly like the frog: from 0 it moves to 1 with certainty, otherwise it is fair.

Step 2. Reaching pad 10 is the same as S_n hitting $+10$ *or* -10, starting from 0.

Step 3. Fair two-barrier duration, with $k = 10$ and $N = 20$ (shifting so that the barriers are at 0 and 20): $k(N - k) = 10 \cdot 10 = 100$.

Step 4. Check by recurrence: $D_k = 100 - k^2$ satisfies $D_k = 1 + \frac{1}{2}(D_{k-1} + D_{k+1})$ for $1 \le k \le 9$, $D_0 = 1 + D_1$ and $D_{10} = 0$. In particular, $D_0 = 100$, and $D_5 = 75$.

Answer. 100 hops (in general N^2).

Analogy – make it intuitive

A ball bouncing inside a corridor with a springy wall at one end and an exit at the other. Instead of reflecting the ball, *unfold the corridor*: the wall disappears and the corridor doubles in length. Reflecting barriers are just absorbing ones on a mirror-image track.

Example 16 | A frog on a 12-hour clock

A frog sits at the "12" of a clock face. Each second it hops to an adjacent hour (clockwise or counter-clockwise, probability $\frac{1}{2}$ each). (a) What is the expected time to first reach "6"? (b) What is the expected time to have visited *every* hour (the cover time)?

Step-by-step solution

Step 1. (a) Unwrap the clock: hitting "6" from "12" means the unwrapped walk first reaches $+6$ or -6, with start 0. This is fair gambler's ruin on $[-6, 6]$: $k = 6$, $N = 12$, so $D = 6 \cdot 6 = 36$.

Step 2. (b) Suppose j distinct hours have been visited ($1 \le j \le 11$). They form a contiguous arc, and the frog sits at one end of it, right next to a new hour on that side.

Step 3. The next new hour is whichever of the two hours bordering the arc is reached first. This is exit from an interval of length $j + 1$ starting one step from an end: the expected time is $1 \cdot j = j$.

Step 4. Summing: $\sum_{j=1}^{11} j = \dfrac{11 \cdot 12}{2} = 66$. In general the cover time of an N-cycle is $\dfrac{N(N-1)}{2}$.

Answer. (a) 36 seconds. (b) 66 seconds.

Analogy – make it intuitive

Painting a fence with a drunk painter. Having painted j boards in a row, he wobbles around in the painted section until he stumbles onto a fresh board at either edge; the j-th new board costs j steps on average. It gets slower as the painted section grows.

Example 17 | Expected visits to a state before absorption

A fair walk on $\{0, \dots, 10\}$ with absorbing ends starts at 3. What is the expected number of times it is at position 5 before absorption?

Step-by-step solution

Step 1. *Reach 5?* From 3, hit 5 before 0: probability $\frac{3}{5}$ (fair gambler's ruin on $[0, 5]$).

Step 2. *Once at 5, how many visits?* Visits are geometric. The chance of *escaping* (reaching 0 or 10 before returning to 5) is: step right ($\frac{1}{2}$) then hit 10 before 5 ($\frac{1}{5}$), or step left ($\frac{1}{2}$) then hit 0 before 5 ($\frac{1}{5}$). So

$$\text{escape} = \tfrac{1}{2} \cdot \tfrac{1}{5} + \tfrac{1}{2} \cdot \tfrac{1}{5} = \tfrac{1}{5},$$

and the expected number of visits from 5 is $1/\text{escape} = 5$.

Step 3. Multiply: $\mathbb{E}[\text{visits to } 5] = \frac{3}{5} \cdot 5 = 3$.

Step 4. General formula (fair walk, absorbing at 0 and N): $G(k, j) = \dfrac{2 \min(k, j)\,(N - \max(k, j))}{N}$. Here $\dfrac{2 \cdot 3 \cdot 5}{10} = 3$ ✓.

Answer. 3 expected visits.

Competition tip

The method "probability of reaching j" × "1/escape probability" is a competition staple: it turns an apparently tough summation into two gambler's-ruin one-liners.

Example 18 | Climb before you return

A fair walk starts at 0. (a) What is the probability it reaches $+5$ before returning to 0? (b) What is the probability it reaches $+5$ or -5 before returning to 0? (c) What is the expected number of visits to 0 (including the start) before $|S_n|$ first hits 5?

Step-by-step solution

Step 1. (a) The first step must be $+1$ (probability $\frac{1}{2}$). From 1, reaching 5 before 0 has probability $\frac{1}{5}$. So $\mathbb{P} = \frac{1}{2} \cdot \frac{1}{5} = \frac{1}{10}$.

Step 2. (b) By symmetry, double it: $\frac{1}{5}$ (or $1/a$ in general).

Step 3. (c) Each "excursion" from 0 ends the process with probability $\frac{1}{5}$ and otherwise returns to 0. The number of visits is geometric with success chance $\frac{1}{5}$, so the mean is 5 (in general, a).

Step 4. Check with Example 16: shift the interval $[-5, 5]$ to $[0, 10]$ and start at the centre ($k = j = 5$, $N = 10$): $G = \frac{2 \cdot 5 \cdot 5}{10} = 5$ ✓.

Answer. (a) $\frac{1}{10}$, (b) $\frac{1}{5}$, (c) 5 visits.

Part E Path Counting: Reflection, Catalan, Ballot and Returns

Example 19 | Counting paths of a fair walk

A fair coin is flipped 10 times; let $S_{10} = \#\text{H} - \#\text{T}$. Find (a) $\mathbb{P}(S_{10} = 2)$, (b) $\mathbb{P}(S_{10} = 0)$, (c) $\mathbb{P}(|S_{10}| \le 2)$.

Step-by-step solution

Step 1. A path ending at j has u up-steps with $u - (10-u) = j$, i.e. $u = \frac{10+j}{2}$.

Step 2. (a) $j = 2 \Rightarrow u = 6$: $\dfrac{\binom{10}{6}}{2^{10}} = \dfrac{210}{1024} \approx 0.205$.

Step 3. (b) $j = 0 \Rightarrow u = 5$: $\dfrac{\binom{10}{5}}{1024} = \dfrac{252}{1024} = \dfrac{63}{256} \approx 0.246$.

Step 4. (c) $|S_{10}| \le 2$ means $S_{10} \in \{-2, 0, 2\}$: $\dfrac{210 + 252 + 210}{1024} = \dfrac{672}{1024} = \dfrac{21}{32} \approx 0.656$.

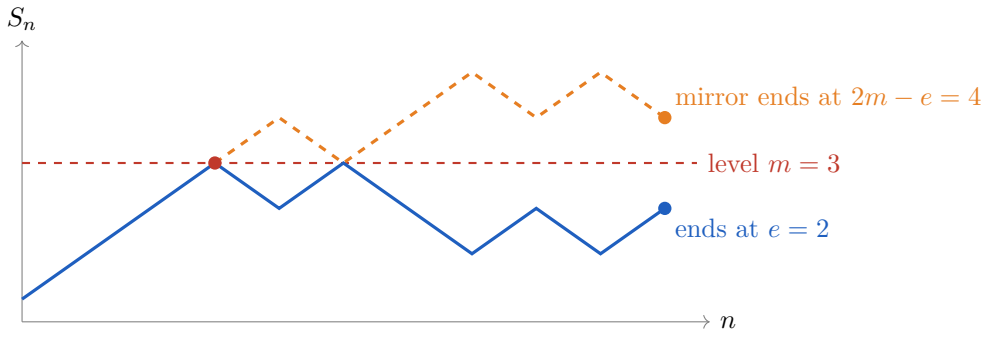
Answer. (a) $\frac{105}{512}$ (b) $\frac{63}{256}$ (c) $\frac{21}{32}$.

Analogy – make it intuitive

A Galton board (pachinko): a ball bounces left/right at each peg. The number of routes to a slot is Pascal's triangle, and the chance of the ball ending in a slot is its route count divided by 2^{rows}.

Example 20 | Reflection principle — the running maximum

A fair walk makes 10 steps from 0. What is the probability that it reaches +3 at some time?



Step-by-step solution

Step 1. Parity: S_{10} is even, so "end ≥ 3" means "end ≥ 4". Every path ending at ≥ 4 certainly touched 3.

Step 2. Paths ending at $e < 3$ (i.e. $e \le 2$) that touch 3: reflect the part after the *first* touch in the line $y = 3$. This maps them bijectively to paths ending at $6 - e \ge 4$ (picture above).

Step 3. Hence $\#\{\text{touch } 3\} = \#\{\text{end} \ge 4\} + \#\{\text{end} \ge 4\} = 2\,\#\{\text{end} \ge 4\}$.

Step 4. Count paths ending at $4, 6, 8, 10$: $\binom{10}{7} + \binom{10}{8} + \binom{10}{9} + \binom{10}{10} = 120 + 45 + 10 + 1 = 176$.

Step 5. $\mathbb{P} = \dfrac{2 \cdot 176}{1024} = \dfrac{352}{1024} = \dfrac{11}{32} \approx 0.344$.

Answer. $\dfrac{11}{32}$. General rule: $\mathbb{P}(\max_{i \le n} S_i \ge m) = 2\mathbb{P}(S_n > m) + \mathbb{P}(S_n = m)$.

Analogy – make it intuitive

A mirror laid along the barrier. As soon as a path hits the mirror, switch to following its reflection. Every bad path now corresponds to exactly one path that *ends beyond* the barrier, and those are easy to count.

Example 21 | Catalan numbers: staying above the axis

(a) How many sequences of 4 up-steps and 4 down-steps never go below the starting level? (b) Equivalently, how many valid strings of 4 pairs of parentheses are there? (c) If a sequence with 4 ups and 4 downs is chosen at random among *all* $\binom{8}{4}$, what is the probability it never goes below 0?

Step-by-step solution

Step 1. Total sequences with 4 ups and 4 downs: $\binom{8}{4} = 70$.

Step 2. Bad sequences touch -1. Reflect the part up to the first touch of -1: these correspond to paths from 0 to -2 with 8 steps, i.e. 3 ups and 5 downs: $\binom{8}{3} = 56$.

Step 3. Good sequences: $70 - 56 = 14 = C_4 = \frac{1}{5}\binom{8}{4}$. Parentheses: each "(" is an up-step and each ")" a down-step, so the answer to (b) is also 14.

Step 4. (c) $\dfrac{14}{70} = \dfrac{1}{5}$.

Answer. (a),(b) $14 = C_4$. (c) $\dfrac{1}{5}$.

Competition tip

Memorise $C_0, \ldots, C_6 = 1, 1, 2, 5, 14, 42, 132$. Catalan numbers count balanced parentheses, Dyck paths, non-crossing chord matchings, binary trees, triangulations, and ballot sequences.

Example 22 | Mosteller #22 — The ballot box

In an election Albert gets a votes and Benjamin gets b votes, $a > b$. Ballots are counted in random order. What is the probability that at some time after the first tally the candidates are tied? Evaluate for $(a, b) = (3, 2)$ and $(60, 40)$.

Step-by-step solution

Step 1. Condition on the first ballot. If it is for Benjamin (probability $\frac{b}{a+b}$), the lead flips at some point because Albert finishes ahead, so a tie is *certain*.

Step 2. If the first ballot is for Albert and a tie ever occurs, reflect the portion up to the first tie (swap A and B). This maps "A-first-with-a-tie" paths bijectively to "B-first" paths.

Step 3. So $\mathbb{P}(\text{tie}) = \dfrac{b}{a+b} + \dfrac{b}{a+b} = \dfrac{2b}{a+b}$.

Step 4. $(3, 2)$: $\frac{4}{5}$ (8 of the 10 sequences tie). $(60, 40)$: $\frac{80}{100} = \frac{4}{5}$. Only the ratio a/b matters.

Step 5. Equivalently, $\mathbb{P}(\text{no tie, A always strictly ahead}) = \dfrac{a-b}{a+b}$.

Answer. $\dfrac{2b}{a+b}$; both cases give $\frac{4}{5}$.

Analogy – make it intuitive

A race between a hare and a tortoise on a long road. If the tortoise is ahead at the start (first ballot for B), the hare *must* later catch up to finish ahead. The mirror trick says that "hare starts ahead but is caught later" is exactly as likely as "tortoise starts ahead".

Example 23 | Ballot problem with "never behind"

Candidate A gets 5 votes and B gets 3, counted in random order. What is the probability A is *never behind* (ties allowed) at any stage?

Step-by-step solution

Step 1. Total orderings: $\binom{8}{3} = 56$.

Step 2. Bad orderings put A behind at some time, i.e. the walk touches -1. Reflect the part after the first touch of -1 in the line $y = -1$: the final value $+2$ becomes $-1 - (2+1) = -4$. So bad paths $\leftrightarrow$ paths ending at -4.

Step 3. With 8 steps and end -4 we need $u - d = -4$, $u + d = 8$, so $u = 2$: $\binom{8}{2} = 28$ bad orderings.

Step 4. Good: $56 - 28 = 28$, so $\mathbb{P} = \frac{28}{56} = \frac{1}{2}$.

Step 5. Formula check: $\dfrac{a-b+1}{a+1} = \dfrac{5-3+1}{6} = \dfrac{1}{2}$ ✓.

Answer. $\dfrac{1}{2}$.

Example 24 | Mosteller #23 — Ties in matching pennies

Players A and B match pennies N times, keeping a tally of gains and losses. After the first toss, what is the probability that at no time are they even? Evaluate for $N = 4$, $N = 10$, and $N = 11$.

Step-by-step solution

Step 1. Let S_k be A's net gain after k tosses. "Never even after the first toss" means $S_k \neq 0$ for $k = 1, \ldots, N$.

Step 2. Result: for $N = 2n$, $\mathbb{P}(\text{no tie}) = \dfrac{\binom{2n}{n}}{2^{2n}} = u_{2n}$, and for $N = 2n+1$ the same value as $N = 2n$.

Step 3. $N = 4$: $\dfrac{\binom{4}{2}}{16} = \dfrac{6}{16} = \dfrac{3}{8}$ (matches the book's list with 6 starred sequences out of 16).

Step 4. $N = 10$: $\dfrac{\binom{10}{5}}{1024} = \dfrac{252}{1024} = \dfrac{63}{256} \approx 0.246$.

Step 5. $N = 11$: same as $N = 10$, so $\dfrac{63}{256}$. (The 11th toss cannot create a tie from $S_{10} \neq 0$, because S_{10} is even and non-zero.)

Answer. $\frac{3}{8}$, $\frac{63}{256}$, $\frac{63}{256}$.

Common trap

Intuition says that after a long game a tie "should" have happened. But with $N = 10$ there is still a 25% chance the walk never returns. The no-return probability decays only like $1/\sqrt{\pi n}$,

which is very slow.

Example 25 | Never behind in 20 flips

A fair coin is flipped 20 times. What is the probability that the number of heads is *never less than* the number of tails at any point?

Step-by-step solution

Step 1. We need $S_k \geq 0$ for all $k \leq 20$. Condition on the final value $e \in \{0, 2, \ldots, 20\}$.

Step 2. Paths ending at e that never go below 0: (all paths ending at e) minus (those touching -1). Reflecting across -1, the bad ones biject with paths ending at $-e-2$, which are as numerous as paths ending at $e+2$. So the count is $N(e) - N(e+2)$ where $N(j) = \binom{20}{(20+j)/2}$ (and $N(22) = 0$).

Step 3. Sum over $e = 0, 2, \ldots, 20$: the sum telescopes to $N(0) - N(22) = N(0) = \binom{20}{10}$.

Step 4. $\mathbb{P} = \dfrac{\binom{20}{10}}{2^{20}} = \dfrac{184756}{1048576} \approx 0.176.$

Answer. $u_{20} = \dfrac{\binom{20}{10}}{2^{20}} \approx 0.176.$

Competition tip

Memorise the equivalence $\mathbb{P}(S_k \geq 0\ \forall k \leq 2n) = \mathbb{P}(S_{2n} = 0) = \mathbb{P}(S_k \neq 0\ \forall\, 1 \leq k \leq 2n) = \binom{2n}{n}4^{-n}$. Three different questions, one number.

Example 26 | When does the walk first return to 0?

A fair walk starts at 0. (a) Find the probability that its *first* return to 0 occurs at time 8. (b) Find the probability that it has *not* returned to 0 by time 8.

Step-by-step solution

Step 1. Let $u_{2n} = \binom{2n}{n}4^{-n}$. The first return time f_{2n} satisfies $f_{2n} = u_{2n-2} - u_{2n} = \dfrac{u_{2n}}{2n-1}$.

Step 2. Values: $u_6 = \frac{20}{64} = \frac{5}{16}$ and $u_8 = \frac{70}{256} = \frac{35}{128}$.

Step 3. (a) $f_8 = u_6 - u_8 = \frac{80}{256} - \frac{70}{256} = \frac{10}{256} = \frac{5}{128}$. (Check: $u_8/7 = \frac{35}{128\cdot 7} = \frac{5}{128}$ ✓.)

Step 4. (b) No return by time 8: $u_8 = \frac{35}{128} \approx 0.273$.

Step 5. Consistency: $f_2 + f_4 + f_6 + f_8 = \frac{1}{2} + \frac{1}{8} + \frac{1}{16} + \frac{5}{128} = \frac{93}{128} = 1 - \frac{35}{128}$ ✓.

Answer. (a) $\frac{5}{128}$. (b) $\frac{35}{128} \approx 0.273$.

Part F Random Walks in Two and Three Dimensions

Example 27 | Mosteller #51 — Two-dimensional random walk

A particle starts at the origin of the integer lattice and at each move goes one step north-east, north-west, south-east or south-west with probability $\frac{1}{4}$ each. (a) Find the probability it is at the origin after 2, 4 and 6 moves. (b) Does it return to the origin with probability 1?

Step-by-step solution

Step 1. Each move changes x by ± 1 and y by ± 1, with the two choices *independent* fair coins. So x and y are independent 1D fair walks.

Step 2. Hence $\mathbb{P}(\text{at origin at } 2n) = \left[\binom{2n}{n}4^{-n}\right]^2 = u_{2n}^2$.

Step 3. (a) $n=1$: $(\frac{1}{2})^2 = \frac{1}{4}$. $n=2$: $(\frac{3}{8})^2 = \frac{9}{64}$. $n=3$: $(\frac{5}{16})^2 = \frac{25}{256}$.

Step 4. (b) For large n, $u_{2n}^2 \approx \dfrac{1}{\pi n}$. The expected number of returns is $\sum_n \frac{1}{\pi n}$, a harmonic series that diverges.

Step 5. The number of returns is geometric: if the return probability were $P < 1$, the mean number of returns would be $\frac{P}{1-P} < \infty$. Since the mean is infinite, $P = 1$.

Answer. (a) $\frac{1}{4}$, $\frac{9}{64}$, $\frac{25}{256}$. (b) Yes, the 2D walk is recurrent (return probability 1).

Analogy – make it intuitive

A drunk man in an infinite city grid: he will eventually stumble back to the bar. But it may take a very long time, because the expected return time is infinite.

Example 28 | Mosteller #52 — Three-dimensional random walk

A particle starts at the origin of 3-space. At each move each of the 3 coordinates independently changes by ± 1 (so it moves to one of the 8 corners of a cube). (a) Find $\mathbb{P}(\text{at origin at time } 2n)$ for $n = 1, 2, 3$. (b) Estimate the probability that it ever returns to the origin. (c) How does this compare with the 6-neighbour walk on the integer lattice?

Step-by-step solution

Step 1. Independent coordinates: $\mathbb{P}(\text{at origin at } 2n) = u_{2n}^3$. For $n = 1, 2, 3$: $(\frac{1}{2})^3 = \frac{1}{8} = 0.125$; $(\frac{3}{8})^3 \approx 0.0527$; $(\frac{5}{16})^3 \approx 0.0305$.

Step 2. For large n, $u_{2n}^3 \approx (\pi n)^{-3/2}$. Now $\sum n^{-3/2}$ *converges*, so the expected number of returns μ is finite.

Step 3. Numerics: the partial sum for $n \le 6$ is ≈ 0.255 and the tail $\sum_{n\ge 7}(\pi n)^{-3/2}$ adds ≈ 0.138, giving $\mu \approx 0.39$.

Step 4. The number of returns is geometric, so $\mu = \dfrac{P}{1-P}$, i.e. $P = \dfrac{\mu}{1+\mu} \approx \dfrac{0.393}{1.393} \approx 0.28$.

Step 5. (c) For the usual 6-neighbour lattice walk (Pólya's constant) the return probability is ≈ 0.34. Either way, the 3D walk is **transient**: with positive probability it never comes back.

Answer. (a) 0.125, 0.0527, 0.0305. (b) $P \approx 0.28$ (strictly less than 1). (c) ≈ 0.34 for the 6-neighbour walk.

Common trap

Textbook note. Mosteller's printed estimate for this walk is $\mu \approx 0.315$ and $P \approx 0.239$. Summing the series more carefully gives $\mu \approx 0.393$, $P \approx 0.282$. The qualitative conclusion ($P < 1$) is the same, so don't memorise the decimal digits, just the idea: $\sum 1/n$ *diverges (2D returns),* $\sum n^{-3/2}$ *converges (3D escapes).*

Analogy – make it intuitive

Kakutani's drunk bird. On the ground (1D or 2D) there are so few directions that you keep bumping into the same places, so you eventually find home. In the air (3D) there is so much more room that the bird has a positive chance of drifting away forever.

Part G Rapid-Fire Drills and Cheat Sheet

Rapid-fire drills (cover the answers!)

#	Question	Answer
1	Fair game, start \$7, target \$20. Win probability and expected number of rounds?	$\frac{7}{20}$; $7 \cdot 13 = 91$
2	$p = \frac{2}{3}$ up, unlimited bank, start at 1: probability of ever hitting 0?	$\frac{1}{2}$
3	Ballot with $a = 9$, $b = 6$: probability A is strictly ahead throughout?	$\frac{3}{15} = \frac{1}{5}$
4	$\mathbb{P}(S_{12} = 0)$ for a fair walk?	$\frac{924}{4096} = \frac{231}{1024}$
5	Number of balanced parenthesis strings with 3 pairs?	$C_3 = 5$
6	Win/lose/tie probabilities 0.4/0.4/0.2, start \$3, target \$12: win probability and expected rounds?	$\frac{1}{4}$; $\frac{3 \cdot 9}{0.8} = 33.75$
7	Fair walk reflecting at 0, absorbing at 8, start 0: expected steps?	64
8	$p = 0.4$ up, $q = 0.6$ down. Expected all-time maximum and $\mathbb{P}$(ever reach $+2$)?	$\frac{p}{q-p} = 2$; $\left(\frac{2}{3}\right)^2 = \frac{4}{9}$
9	20-cycle: expected time to reach the antipodal vertex, and cover time?	$10 \cdot 10 = 100$; $\frac{20 \cdot 19}{2} = 190$
10	2D walk (4 diagonal moves): $\mathbb{P}$(at origin at step 6)?	$\left(\frac{20}{64}\right)^2 = \frac{25}{256}$

One-page cheat sheet

Gambler's ruin (fortune k, target N, $r = q/p$):

$$W_k = \frac{k}{N} \ (p = q), \qquad W_k = \frac{1 - r^k}{1 - r^N} \ (p \neq q), \qquad W_k = \frac{1}{1 + r^k} \text{ if } N = 2k.$$

Duration: $D_k = k(N - k)/\sigma^2$ (fair; σ^2 = variance per round, $= 1$ for ± 1 steps). Biased: $D_k = \dfrac{NW_k - k}{p - q}$.

Infinite bank: ruin probability $\min\{1, (q/p)^k\}$. Expected max of a downhill walk: $\dfrac{p}{q - p}$. Time

to climb a with drift $p-q>0$: $\dfrac{a}{p-q}$.
Reflecting barrier at 0, absorbing at N (fair): from 0, $D=N^2$ (unfold with $|S_n|$). **Cycle of N:** hit a vertex at distance d in $d(N-d)$; cover time $\frac{N(N-1)}{2}$.
Fair walk path facts: $\mathbb{P}(S_{2n}=0)=u_{2n}=\binom{2n}{n}4^{-n}$; $\mathbb{P}(\text{no return by } 2n)=\mathbb{P}(\text{never negative by } 2n)=u_{2n}$; first return at $2n$: $\dfrac{u_{2n}}{2n-1}$; $\mathbb{P}(\max_{i\le n} S_i \ge m)=2\mathbb{P}(S_n>m)+\mathbb{P}(S_n=m)$.
Ballot: strictly ahead $\frac{a-b}{a+b}$; never behind $\frac{a-b+1}{a+1}$; some tie $\frac{2b}{a+b}$. **Catalan:** $C_n=\frac{1}{n+1}\binom{2n}{n}$.
Visits: (fair, absorbing $0,N$) expected visits to j from k is $\dfrac{2\min(k,j)(N-\max(k,j))}{N}$. Geometric trick: $\mathbb{E}[\#\text{visits}]=1/\text{escape probability}$.
Dimension: $\mathbb{P}(\text{origin at } 2n)=u_{2n}^d\sim(\pi n)^{-d/2}$; $\sum$ diverges for $d=1,2$ (recurrent), converges for $d\ge 3$ (transient).

Five-step competition routine

1. **Draw the line:** mark the start, the absorbing ends, and the step probabilities.
2. **Fair or biased?** Fair $\Rightarrow$ linear/quadratic answers; biased $\Rightarrow$ geometric $r=q/p$.
3. **Look for the shortcut:** halfway trick, martingale (S_n or r^{S_n}), mirror/unfolding, or "ties are pauses".
4. **Unusual boundary?** Reflect it away (unfold) or write one first-step equation per state.
5. **Sanity-check:** do the probabilities lie in $[0,1]$, does a fair game keep expected wealth fixed, do the limits $p\to\frac{1}{2}$ match k/N?